

FUNCTIONS - HOMEWORK

1. Let $A = \{0, 2, 4, 6\}$ and $B = \{1, 3, 5, 7\}$. Determine which of the following relations between A and B forms a function with domain A and codomain B .

- (a) $\{(6, 3), (2, 1), (0, 3), (4, 5)\}$
- (b) $\{(2, 3), (4, 7), (0, 1), (6, 5)\}$
- (c) $\{(2, 1), (6, 3), (4, 5)\}$
- (d) $\{(6, 1), (4, 1), (0, 3), (2, 5), (0, 7)\}$

For those which are functions, determine which are injective and which are surjective.

2. Let $A = \{0, 2, 4, 6\}$ and $B = \{1, 3, 5, 7\}$. Determine which of the following relations between A and B forms a function with domain A and codomain B . For those which are functions, determine which are injective and which are surjective and find their range.

- (a) $R = \{(x, y) : x < y\}$
- (b) $R = \{(x, y) : 2x = y\}$

3. Let $A = \{1, 2, 3\}$ and $B = \{3, 6, 9\}$. Determine which of the following relations between A and B forms a function with domain A and codomain B . For those which are functions, determine which are injective and which are surjective and find their range.

- (a) $R = \{(x, y) : x + 3 = y\}$
- (b) $R = \{(x, y) : 3x = y\}$
- (c) $R = \{(x, y) : x + y \text{ is even and } x + y \geq 8\}$
- (d) $R = \{(x, y) : x + y \text{ is odd and } x + y \leq 10\}$

4. Each of the following defines a function with domain and codomain equal to $\mathbb{Z}$. In each case decide whether or not the function is injective or surjective (or both).

- (a) $f(n) = 2n + 1$
- (b)

$$g(n) = \begin{cases} \frac{1}{2}n & \text{if } n \text{ is even} \\ 2n & \text{if } n \text{ is odd} \end{cases}$$

- (c)

$$h(n) = \begin{cases} n + 1 & \text{if } n \text{ is even} \\ n - 1 & \text{if } n \text{ is odd} \end{cases}$$

5. Determine whether the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ given by

$$f(x) = \begin{cases} x + 2 & \text{if } x \geq 2 \\ 3x - 2 & \text{if } x < 2 \end{cases}$$

is injective or surjective.

6. Determine whether the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} x + 2 & \text{if } x \geq 3 \\ 3x - 2 & \text{if } x < 3 \end{cases}$$

is injective or surjective.

7. State the range of each of the following functions and determine whether each of them is injective or surjective (or both).

(a) $f : \mathbb{Z} \rightarrow \mathbb{N}$ given by $f(x) = x^2 + 1$

(b) $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = 5x - 1$

(c) $h : \mathbb{R} \rightarrow \mathbb{R}$ given by $h(x) = x + |x|$

(d) $k : \mathbb{R} \rightarrow \mathbb{R}$ given by $k(x) = 2x - |x|$

8. The **floor function** assigns to a real number x the largest integer that is less than or equal to x . This is denoted by $\lfloor x \rfloor$.

(a) Let $A = \{-1, 0, 1, 2\}$ and $f : A \rightarrow \mathbb{Z}$ be given by

$$f(x) = \left\lfloor \frac{x^2 + 1}{3} \right\rfloor$$

Find the range of f . Determine if f is injective or surjective (or both).

(b) Determine whether the function $g : \mathbb{Z} \rightarrow \mathbb{Z}$ given by:

$$g(n) = \left\lfloor \frac{n}{2} \right\rfloor$$

is injective or surjective (or both).

9. Let $A = \{x : x \in \mathbb{R} \text{ and } x \neq 0\}$ and $B = \{x : x \in \mathbb{R} \text{ and } x \neq 1\}$. Let the function $f : A \rightarrow B$ be given by

$$f(x) = 1 + \frac{2}{x}$$

Show that f is bijective and determine its inverse function.

10. Let $A = \{x : x \in \mathbb{R} \text{ and } x \neq 5\}$ and $B = \{x : x \in \mathbb{R} \text{ and } x \neq 3\}$. Let the function $f : A \rightarrow B$ be given by

$$f(x) = \frac{3x - 2}{x - 5}$$

Show that f is bijective and determine its inverse function.

11. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x + 3$ and the function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = x^2 - 3x$. Calculate $g \circ f$, $f \circ g$, $f \circ f$ and $g \circ g$.
12. (a) How many times must a single die be rolled to guarantee that some number is obtained twice?
- (b) How many times must a single die be rolled to guarantee that some number is obtained six times?
- (c) How many times must two dice be rolled to guarantee that the same total score is obtained at least twice?
- (d) How many times must two dice be rolled to guarantee that the same total score is obtained at least five time?
13. (a) How many cards must be drawn from a standard pack of 52 cards to guarantee that at least two are the same suit?
- (b) How many cards must be drawn from a standard pack of 52 cards to guarantee that at least four are the same suit?
14. (a) How many people must be selected at random to guarantee that at least 5 of them have a birthday in the same month of the year?
- (b) How many people must be selected at random to guarantee that at least 3 of them have a birthday on the same date (Do not take leap years into account)?
15. Let $S = \{1, 2, 3, \dots, 20\}$.
- (a) How many numbers must be selected from S so that at least two of the numbers selected sum to 21?
- (b) How many even numbers must be selected from S so that at least two of the numbers selected sum to 22?